



Right Heart Dysfunction and Postoperative Renal Injury: Venous Congestion, Renal Perfusion Pressure, and Perioperative Implications

Aaron Siegman¹ · Prabdit Singh Sidhu¹ · David Li¹

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Abstract

Purpose of Review Postoperative acute kidney injury (AKI) remains a leading cause of morbidity after cardiac surgery. This review examines right ventricular (RV) dysfunction and systemic venous congestion as primary, underrecognized mediators of perioperative renal injury, operating through mechanisms largely independent of arterial hypoperfusion.

Recent Findings Human hemodynamic studies consistently demonstrate that central venous pressure is a stronger predictor of renal impairment than cardiac index across cardiovascular disease populations. Intraoperative venous congestion independently predicts postoperative AKI in cardiac surgery, and dynamic intraoperative deterioration of RV function is among the strongest predictors of cardiac surgery-associated AKI identified to date. Elevated renal venous pressure erodes the renal perfusion pressure gradient, producing congestive nephropathy — a potentially reversible renal dysfunction phenotype distinct from ischemic injury.

Summary RV dysfunction and venous congestion should be recognized as primary mediators of perioperative AKI. Postoperative oliguria with elevated venous pressure warrants decongestion rather than fluid administration. Preoperative RV assessment, intraoperative CVP surveillance, and bedside venous congestion phenotyping using tools such as Venous Excess Ultrasound represent practical and immediately applicable perioperative strategies.

Keywords Right ventricular dysfunction · Venous congestion · Acute kidney injury · Cardiorenal syndrome · Perioperative medicine · Cardiac surgery

Introduction

Postoperative acute kidney injury (AKI) is among the most frequent and morbid complications in perioperative medicine. After cardiac surgery, the reported incidence approaches 25% to 30%, and even modest postoperative creatinine elevations are independently associated with prolonged hospitalization, infectious complications, and increased mortality [1–3]. Renal dysfunction in cardiovascular disease is not

purely a comorbidity marker; it is a powerful independent predictor of adverse outcomes [4].

Conventional perioperative AKI models emphasize hypotension, ischemia-reperfusion, inflammation, nephrotoxin exposure, and low cardiac output as key mechanisms precipitating the injury. However, these mechanisms remain important, but do not fully account for the renal dysfunction seen across the perioperative spectrum. A more complete model has emerged from heart failure, pulmonary hypertension, and cardiac surgery literatures. This model emphasizes right heart dysfunction and systemic venous congestion as primary determinants of renal injury, operating through a hemodynamic pathway that is largely independent of arterial hypoperfusion. In multiple clinical cohorts, renal dysfunction correlates more strongly with elevated right-sided filling pressures and central venous pressure (CVP) than with cardiac index or mean arterial pressure [5–7].

This framework is grounded in the modern classification of cardiorenal syndrome (CRS) proposed by Ronco et al. and expanded in the American Heart Association Scientific

✉ David Li
dpli@health.ucdavis.edu

Aaron Siegman
adsiegman@health.ucdavis.edu

Prabdit Singh Sidhu
psidhu@health.ucdavis.edu

¹ Department of Anesthesiology and Pain Medicine, UC Davis Health, Patient Support Services Building 4150 V St, Sacramento, CA 95817, USA

Statement by Rangaswami et al. [4, 8]. Type 1 CRS, defined as acute cardiac dysfunction precipitating AKI, is precisely the scenario that perioperative right heart dysfunction represents. The perioperative setting is uniquely hazardous for the right ventricle (RV) as positive-pressure ventilation, cardiopulmonary bypass (CPB), pulmonary vasoconstriction, arrhythmia, and ischemia-reperfusion can all worsen RV performance, elevate right atrial pressure, and produce renal venous hypertension that is invisible when clinical attention is focused on arterial blood pressure alone [9–11]. This review combines the physiologic and clinical evidence linking right heart dysfunction to postoperative renal injury, with emphasis on venous congestion, renal perfusion pressure, and practical perioperative implications.

Right Ventricular Physiology and Its Relevance to the Kidney

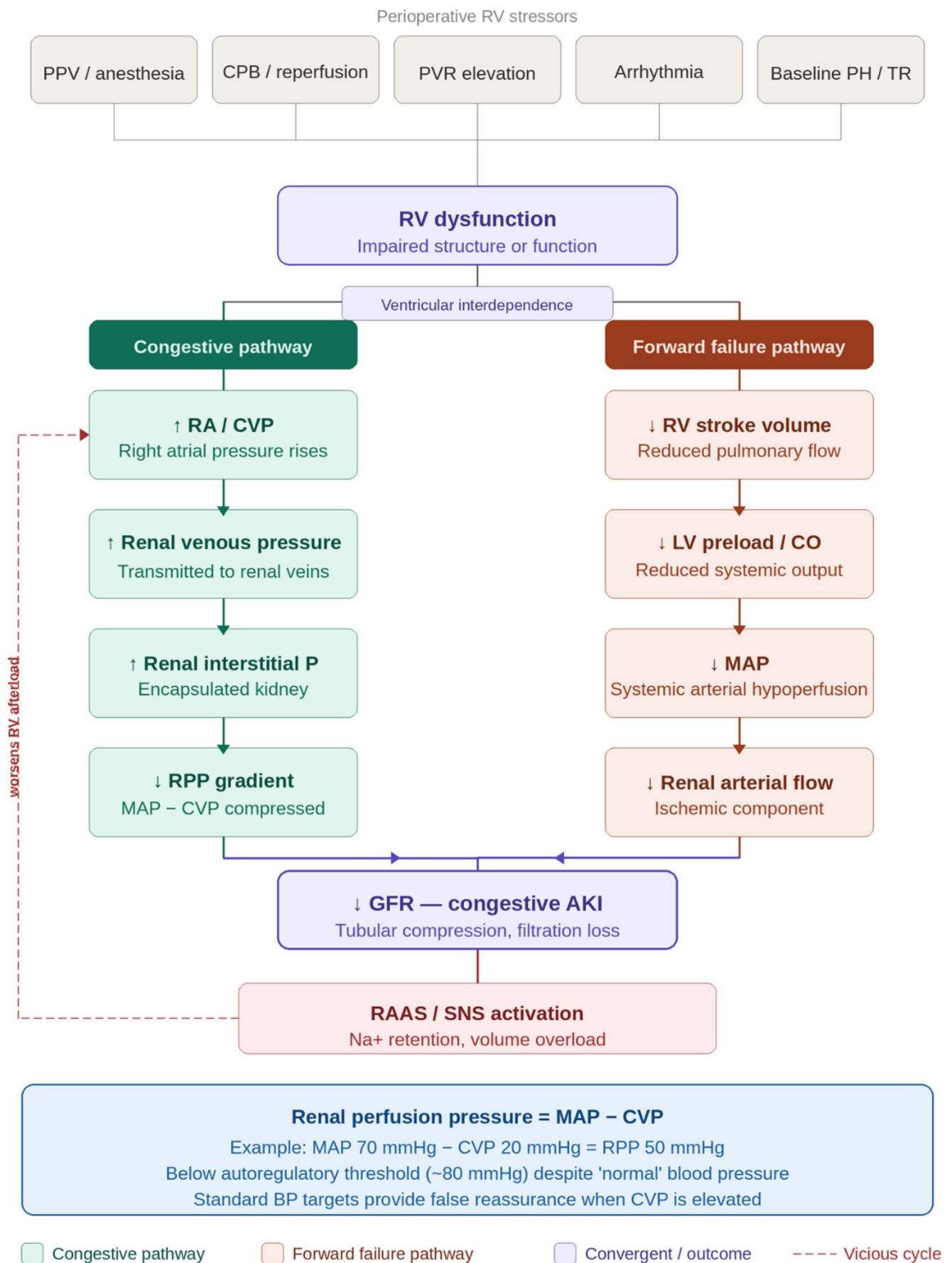
The RV is structurally and functionally distinct from the left ventricle (LV). Haddad et al. describe a thin-walled, crescentic chamber relying predominantly on longitudinal shortening, with myocardial mass approximately one-sixth that of the LV [12]. The RV is adapted to volume loading, but acutely sensitive to afterload: even modest elevations in pulmonary vascular resistance (PVR) can markedly reduce stroke volume. The American Heart Association (AHA) Scientific Statement by Konstam et al. characterizes RV failure as a clinical syndrome of impaired forward pulmonary flow combined with systemic venous congestion, spanning a spectrum from subclinical dysfunction to overt low-output failure with multi-organ congestion [13]. Critically, RV dysfunction and RV failure are not synonymous: dysfunction refers to impaired structure or function without clinical failure, whereas failure implies hemodynamic decompensation with organ congestion, including the kidney. This distinction has direct perioperative relevance, because subclinical RV dysfunction may generate clinically meaningful renal venous congestion before overt RV failure is apparent [10].

Perioperative RV stressors are numerous and often simultaneous. Mattei et al. catalogs the specific operative contributors in cardiac surgery: prolonged CPB, ischemia-reperfusion injury, cardioplegia-related myocardial depression, coronary air embolism, protamine-induced pulmonary vasoconstriction, and reperfusion-related pulmonary vascular loading. Of note, the RV free wall is particularly vulnerable to myocardial mal-protection given its reliance on antegrade cardioplegia delivery techniques, which preferentially protect the LV [11]. Positive-pressure ventilation raises mean intrathoracic pressure, increases RV afterload, and reduces venous return. Blood transfusion and surgical re-exploration are both associated with pulmonary vascular

Fig. 1 Mechanisms linking right ventricular (RV) dysfunction to perioperative acute kidney injury (AKI). Perioperative RV stressors (top) precipitate RV dysfunction, which injures the kidney through two interdependent pathways. The congestive pathway (left, teal) transmits elevated right atrial and central venous pressure (CVP) to the renal vasculature, raising renal venous and interstitial pressure and compressing the renal perfusion pressure (RPP) gradient. When RPP falls below the autoregulatory threshold of approximately 80 mmHg, GFR becomes directly pressure-dependent. The forward failure pathway (right, coral) reduces cardiac output and mean arterial pressure (MAP), limiting renal arterial inflow. Ventricular interdependence amplifies both pathways simultaneously. A vicious cycle (dashed red) is established when renal dysfunction activates the renin-angiotensin-aldosterone (RAAS) and sympathetic nervous systems, promoting sodium and water retention that further worsens RV afterload. The callout box illustrates the RPP equation: $\text{MAP } 70 \text{ mmHg} - \text{CVP } 20 \text{ mmHg} = \text{RPP } 50 \text{ mmHg}$, below the renal autoregulatory threshold despite an apparently acceptable blood pressure. Abbreviations: AKI, acute kidney injury; CO, cardiac output; CPB, cardiopulmonary bypass; CVP, central venous pressure; GFR, glomerular filtration rate; LV, left ventricle; MAP, mean arterial pressure; PH, pulmonary hypertension; PPV, positive-pressure ventilation; PVR, pulmonary vascular resistance; RA, right atrium; RAAS, renin-angiotensin-aldosterone system; RPP, renal perfusion pressure; RV, right ventricle; SNS, sympathetic nervous system; TR, tricuspid regurgitation

resistance elevations. Arora et al., writing for the POQI IX Consortium, stress that the operating room is a uniquely hostile environment for patients with baseline RV vulnerability and that transient intraoperative insults can precipitate sustained postoperative RV dysfunction [10]. Merlo et al. reinforce this by demonstrating that preoperative RV dysfunction is independently associated with worse postoperative outcomes across coronary artery bypass grafting (CABG), valvular, transplant, and left ventricular assist device (LVAD) surgery; yet standard surgical risk scores, including Society of Thoracic Surgeons (STS) and EuroSCORE, do not formally incorporate RV functional assessment [14].

RV dysfunction threatens the kidney through two interdependent pathways. The first is reduced forward output, where impaired RV stroke volume reduces pulmonary blood flow, LV preload, and systemic cardiac output, creating conditions for renal hypoperfusion. The historically underappreciated second pathway is venous congestion, where elevated RV diastolic and right atrial pressures are transmitted into the systemic venous circulation, raising CVP and renal venous pressure, which reduces renal perfusion pressure. Ventricular interdependence amplifies both. Acute RV dilation shifts the interventricular septum leftward, increasing pericardial constraint, impairing LV filling, and further reducing systemic output. These two pathways simultaneously depress renal arterial inflow and elevate renal venous outflow pressure, producing a double hemodynamic insult to glomerular filtration that is mechanistically distinct from isolated arterial hypoperfusion [15, 16] (Fig. 1). It is the venous congestion pathway that has been most



systematically overlooked. Mean arterial pressure is usually preserved by neurohormonal compensation or with vasoactive medications, but elevated venous pressure can impair renal function silently and continuously.

Renal Hemodynamics: Why Venous Pressure Matters

Glomerular filtration depends on the renal perfusion pressure gradient (RPP): the difference between mean arterial pressure and renal venous pressure (RVP), where CVP or right atrial pressure are surrogates that can be used for RVP. When CVP rises, RPP is eroded even if arterial pressure appears adequate. Guazzi et al. formalize this in their lung-right heart-kidney model: when RPP falls below the autoregulatory threshold of approximately 80 mmHg, glomerular filtration rate (GFR) becomes directly pressure-dependent, and small additional reductions produce disproportionate filtration loss [16]. A “normal” mean arterial pressure of 65 to 70 mmHg in a patient with a CVP of 20 mmHg may therefore represent an RPP below the autoregulatory threshold, a state in which standard perioperative blood pressure targets provide false reassurance. Neurohormonal compensation through renin-angiotensin-aldosterone and sympathetic activation may preserve arterial pressure while worsening filtration fraction and driving tubular sodium retention, producing oliguria that mimics, but is mechanistically distinct from, volume depletion [4].

Husain-Syed et al. formalize this as congestive nephropathy: a potentially reversible renal dysfunction phenotype driven by impaired renal venous outflow and rising renal interstitial pressure [17]. Because the kidney is encapsulated, congestion rapidly translates into interstitial pressure elevation with compression of peritubular capillaries and tubular segments. A vicious cycle ensues: venous hypertension promotes hormonal activation, hepatic congestion, intra-abdominal hypertension, intestinal ischemia, oxidative stress, and enhanced tubular sodium reabsorption, each of which worsens volume overload and RV afterload. CVP elevations as modest as 6 mmHg above normal are associated with a steep nonlinear GFR decline, placing the renal injury threshold well below cardiogenic shock [6, 17]. Experimental confirmation comes from Rao et al., whose porcine tamponade model demonstrated that decompressing intratubular pressure via renal negative pressure treatment restored GFR and natriuresis without altering systemic hemodynamics, isolating elevated tubular pressure, rather than reduced renal blood flow, as a primary driver of congestive AKI [18]. Boerrigter et al. note that in some heart failure cohorts, CVP is a better correlate of low GFR than

either cardiac index or mean arterial pressure, a finding that directly challenges the reflex interpretation of perioperative oliguria as hypovolemia [19].

Human Clinical Evidence

Mullens et al. provide the landmark human demonstration in a prospective cohort of 145 patients with advanced decompensated heart failure managed with Pulmonary Artery Catheter (PAC)-guided therapy [5]. Worsening renal function (creatinine increase ≥ 0.3 mg/dL) occurred in 40% of patients. CVP was a powerful predictor (receiver operating characteristic (ROC) AUC 0.734, $p < 0.0001$), with 75% of patients with CVP > 24 mmHg developing worsening renal function. Meanwhile, cardiac index was not predictive of renal dysfunction (AUC 0.552, $p = 0.6$). Renal perfusion pressure was similar between those who did and did not develop worsening renal function, dissociating injury from arterial hypoperfusion. Achieving CVP < 8 mmHg was renal protective ($p = 0.01$), further supporting that venous congestion, not cardiac index, was the dominant hemodynamic driver for worsening renal function.

Damman et al. extended this to 2,557 patients across a broad cardiovascular spectrum, including valvular disease, ischemic heart disease, and cardiomyopathy, all undergoing right heart catheterization. They found CVP independently predicted lower eGFR (partial $r = -0.108$, $p < 0.0001$) and long-term all-cause mortality (HR 1.03 per mmHg, 95% CI 1.01–1.05, $p = 0.003$) after full multivariate adjustment [6]. The breadth of diagnoses confirms that the CVP-renal function relationship is not confined to advanced heart failure but extends across the hemodynamic profiles encountered in surgical patients.

Grodin et al. refined the phenotype by examining the ratio of right atrial pressure (RAP) to pulmonary capillary wedge pressure (PCWP) in 367 acutely decompensated heart failure patients. Increasing RAP/PCWP quartile was independently associated with worse eGFR throughout hospitalization and with increased all-cause mortality (HR 2.4, 95% CI 1.4–3.9, adjusted for cardiac index, PVR, and eGFR) [7]. A right-dominant congestion profile, characterized by disproportionate RV relative to LV filling pressure, identifies a particularly adverse cardiorenal phenotype, likely reflecting isolated RV dysfunction or abnormal ventricular interaction. Kormos et al. found the analogous CVP/PCWP ratio > 0.63 to independently predict post-LVAD RV failure (OR 2.3, 95% CI 1.2–4.3), supporting the generalizability of ratio-based RV-renal phenotyping across circulatory failure states [20].

Table 1 Key clinical evidence linking venous congestion and RV dysfunction to renal injury

Study	Population	Key finding(s)	Impact on renal outcome
Mullens et al. [5]	Decompensated Heart Failure	CVP was a stronger predictor of worsening renal function than Cardiac Index.	75% of patients with CVP > 24 mmHg developed renal injury.
Damman et al. [6]	Broad Cardiovascular Spectrum	CVP independently predicted lower eGFR and increased mortality.	Each 1 mmHg increase in CVP raised mortality risk by 3%.
Lopez et al. [22]	Elective Cardiac Surgery	Intraoperative CVP area-under-the-curve (AUC) predicted AKI.	Pre-bypass congestion specifically initiated renal tubular stress.
Tohi et al. [23]	Cardiac Surgery (Post-CPB)	Dynamic deterioration of RV Fractional Area Change (RVFAC).	ΔRVFAC was the strongest independent predictor of AKI (AUC 0.902).

Chen et al. extend these observations to critically ill patients. Among 1,879 intensive care unit (ICU) patients with echocardiography within 24 h of admission, isolated RV dysfunction showed a trend toward increased AKI risk (adjusted RR 1.35, 95% CI 0.90–2.10, $p = 0.14$) that did not reach statistical significance, while biventricular dysfunction was independently associated with a 1.67-fold adjusted risk (95% CI 1.23–2.31, $p = 0.002$) [21]. Strikingly, AKI in the setting of isolated RV dysfunction carried a 7.85-fold hospital mortality increase (95% CI 2.89–21.3), far exceeding the approximately 2-fold mortality associated with isolated LV dysfunction and AKI. Peripheral edema independently modified AKI risk regardless of ventricular function category, confirming that volume overload is a pathophysiologic active renal stressor, not purely an epiphenomenon (Table 1).

Cardiac Surgery Evidence: Intraoperative Venous Congestion and RV Dysfunction as Predictors of AKI

Karkouti et al. established the epidemiologic foundation in 3,500 cardiac surgery patients across seven centers: AKI occurred in 24% and was associated with greater than 4-fold mortality odds across all severity thresholds [2]. Perioperative red blood cell transfusion and surgical re-exploration, both associated with pulmonary vascular loading and RV stress, emerged as modifiable risk factors, connecting standard AKI risk frameworks to the RV-congestion pathway even without direct hemodynamic analysis.

Lopez et al. provide the most direct intraoperative evidence. In 425 patients undergoing elective cardiac surgery with CPB, venous congestion was quantified as the CVP area-under-the-curve above thresholds of 12, 16, and 20 mmHg. AKI by Kidney Disease: Improving Global Outcomes (KDIGO) criteria, defined as a serum creatinine increase of 0.3 mg/dL or greater within 48 h, a 1.5-fold or greater rise from baseline within 7 days, or urine output less than 0.5 mL/kg/hour for 6 or more hours, occurred in 24.5%. Greater intraoperative venous congestion was

independently associated with higher AKI odds at all three thresholds (OR 1.06 to 1.30 per 60 mmHg·min increment, all $p \leq 0.013$) [22]. Pre-bypass venous congestion specifically correlated with elevated urinary tissue inhibitor of metalloproteinase-2 and insulin-like growth factor binding protein-7 (TIMP-2·IGFBP7), a validated early biomarker of renal tubular stress, during and after CPB. The temporal specificity of pre-bypass congestion as the most harmful exposure implies that RV dysfunction and elevated venous pressure before skin incision may initiate a molecular renal stress cascade that predates CPB-related inflammatory injury.

Tohi et al. directly measured intraoperative RV function by transesophageal echocardiography (TEE) before and after CPB in a prospective cohort of 34 cardiac surgery patients, with simultaneous PAC monitoring. RV fractional area change (RVFAC), RV ejection fraction (RVEF), and RV longitudinal strain all declined significantly post-CPB, and CVP increased (7.2 ± 3.5 to 9.7 ± 3.7 mmHg, $p = 0.0001$) [23]. Change in RVFAC was the strongest independent predictor of postoperative AKI (Spearman $\rho = -0.61$, $p < 0.0001$), with an ROC AUC of 0.902 and sensitivity of 92%. Change in LV function was not independently predictive after accounting for RV changes. This study is uniquely important as it identifies dynamic, intraoperative RV deterioration rather than static preoperative status alone as the proximate renal risk, and it mechanistically links the venous congestion exposure documented by Lopez et al. to its upstream driver [22].

Together, these three studies support a coherent perioperative model in which postoperative AKI is mediated not only by hypotension or CPB-related inflammation, but by a sequence of RV dysfunction (whether pre-existing, acutely worsened, or dynamically deteriorating) leading to elevated right-sided pressures, venous congestion, and renal tubular stress. The LVAD literature reinforces this model as well. Brisco et al. demonstrate that early post-LVAD renal improvement, attributable to correction of hemodynamic derangement and volume overload, is frequently followed by late deterioration when venous congestion and RV dysfunction recur.

Table 2 Management strategies to prevent perioperative AKI

Phase	Strategy	Rationale
Intraoperative	Judicious Fluid Use	Prevent RV dilation and septal shift.
	Lung-Protective Vent.	Minimize mean intrathoracic pressure to reduce RV afterload.
	Pulmonary Vasodilators	Use inhaled NO or prostacyclin for acute RV strain.
Postoperative	VExUS Monitoring	Identify renal venous congestion via bedside Doppler.
	Targeted Decongestion	Use diuresis or ultrafiltration if CVP is high and oliguria persists.
	Vasopressor Preference	Favor vasopressors over fluids if CVP > threshold.

Perioperative Implications

Preoperative Assessment

Preoperative identification of RV vulnerability is the necessary foundation of this paradigm. High-risk profiles include established pulmonary hypertension, documented RV dysfunction, significant tricuspid regurgitation, prior cardiorenal syndrome, advanced heart failure, or procedures associated with substantial RV stress, including combined CABG-valve surgery, LVAD implantation, and heart transplantation. Standard surgical risk scores do not formally incorporate RV functional assessment, a gap emphasized by both Merlo et al. and Mattei et al. [11, 14]. Preoperative echocardiographic markers with established prognostic value include RVFAC < 35%, Tricuspid Annular Plane Systolic Excursion (TAPSE) < 17 mm, peak systolic annular velocity S' < 9.5 cm/s, and free wall longitudinal strain less negative than -20%. Arora et al. propose a pragmatic perioperative RV scoring system integrating baseline RV function with anticipated operative stress, which remains a conceptually useful tool even without a kidney-specific outcome focus [10]. Right heart catheterization with assessment of RAP, PCWP, and their ratio should be considered in patients with pulmonary hypertension or advanced heart failure, where the CVP/PCWP or RAP/PCWP phenotype provides both renal risk information and surgical planning guidance [7, 20].

Intraoperative Management

Intraoperative goals in high-risk patients include preservation of sinus rhythm and atrioventricular synchrony, avoidance of hypoxemia, hypercapnia, and acidemia, minimization of unnecessary PVR elevation, and maintenance of adequate RV coronary perfusion pressure [9, 10]. RV preload should be optimized rather than indiscriminately augmented. The reflex administration of fluid for oliguria may be directly harmful when elevated CVP and RV congestion exist as the mechanism rather than volume depletion. Administering additional volume worsens venous pressure, intensifies renal congestion, and risks precipitating RV

dilation with septal shift [19, 21]. The Lopez et al. finding that pre-bypass congestion is the most renal-toxic intraoperative exposure suggests that preoperative hemodynamic optimization, judicious pre-bypass fluid management, and ventilator strategies that minimize mean intrathoracic pressure are potentially modifiable determinants of postoperative AKI [22]. Where arterial pressure is inadequate, vasopressor support should be preferred over volume loading when CVP is already elevated, and pulmonary vasodilators (inhaled nitric oxide, inhaled prostacyclin) should be considered early in patients at high risk for intraoperative RV decompensation (Table 2).

Postoperative Monitoring

Postoperative oliguria or rising creatinine should trigger active assessment for RV dysfunction and venous congestion alongside the standard evaluation for hemorrhage and hypovolemia. As emphasized in perioperative RV management literature, rising CVP with worsening oliguria should raise the differential of RV failure rather than triggering further fluid administration [9, 19]. Useful clinical and monitoring parameters include CVP trends, echocardiographic RV size and function, new or worsening tricuspid regurgitation, inferior vena cava (IVC) dilatation with reduced collapsibility, hepatic venous flow reversal, and fluid balance trajectory. Beaubien-Souligny et al. validated the Venous Excess Ultrasound (VExUS) grading system in 145 cardiac surgery patients: the highest VExUS grade, defined by IVC ≥ 2 cm with severe hepatic, portal, and intrarenal vein Doppler abnormalities, was associated with a 3.69-fold AKI hazard (95% CI 1.65–8.24, adjusted HR 2.82) and outperformed single CVP measurements with a positive likelihood ratio of 6.37. VExUS provides a non-invasive, repeatable tool for bedside congestion phenotyping readily applicable in the cardiac surgery ICU [24]. When significant RV dysfunction and venous congestion are confirmed, treatment priorities should include relief of congestion (diuresis, ultrafiltration if diuretic-resistant), reduction of RV afterload (pulmonary vasodilators, ventilator optimization), and avoidance of further fluid loading, a management approach directly opposite to the traditional response to oliguria (Table 2).

Conclusion

The relationship between right heart dysfunction and postoperative renal injury is best understood through the interaction of three interdependent hemodynamic mechanisms: reduced forward pulmonary output, systemic venous hypertension, and reduction of the renal perfusion pressure gradient. These mechanisms operate independently of and additively with classical ischemic mechanisms of perioperative AKI, and they explain patterns of oliguria and renal dysfunction that cannot be reconciled with hypotension or hypovolemia alone. The concept of congestive nephropathy, a hydraulic and potentially reversible renal dysfunction driven by elevated renal venous pressure and interstitial pressure rather than ischemia, provides the unifying pathophysiologic framework [17]. Human hemodynamic studies, cardiac surgical series, and critical care cohorts consistently demonstrate that venous congestion and RV dysfunction are strongly and independently associated with AKI and mortality [5–7, 21]. Cardiac surgical studies directly demonstrate that intraoperative venous congestion and dynamic deterioration of RV function are among the strongest predictors of postoperative AKI identified to date [22, 23]. For perioperative clinicians, this evidence demands a conceptual reorientation. Postoperative oliguria and rising creatinine should not reflexively prompt fluid administration when the right heart is failing and venous pressure is elevated, because in this setting fluid worsens rather than relieves the renal insult. Clinical assessment must explicitly integrate RV function, CVP trends, and systemic congestion alongside evaluation for hypovolemia and hemorrhage. Preoperative echocardiographic RV assessment, intraoperative CVP surveillance, and postoperative multimodal venous congestion phenotyping using tools such as VExUS represent practical, immediately applicable steps. Important research gaps remain: standardized perioperative RV dysfunction definitions, randomized trials of targeted decongestion as a renal-protective strategy, and formal incorporation of RV parameters into perioperative risk scores are priority research objectives. Recognizing RV dysfunction and systemic venous congestion as primary mediators of perioperative AKI, rather than as consequences of it, changes how we should assess, monitor, and treat some of the most vulnerable patients in perioperative medicine.

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○ This consensus statement from the POQI IX Consortium provides a comprehensive, evidence-based framework for perioperative RV risk stratification, intraoperative management, and escalating treatment of RV failure. It is directly relevant to the clinical recommendations in this review and represents the most current expert guidance available for the perioperative team managing patients with RV vulnerability.

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○ This study demonstrates that preoperative RV dysfunction independently predicts adverse postoperative outcomes across all major cardiac surgical categories, including CABG, valvular surgery, transplantation, and LVAD implantation, yet is not captured by standard risk scores such as STS and EuroSCORE. It provides direct evidence supporting the formal incorporation of RV assessment into perioperative risk stratification.

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○ This review catalogs the specific intraoperative mechanisms by which cardiac surgery precipitates RV dysfunction, including ischemia-reperfusion injury, cardioplegia-related myocardial depression, protamine-induced pulmonary vasoconstriction, and reperfusion loading. It contextualizes the perioperative RV stressor burden that underlies the cardiorenal pathway described in this review.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Human and Animal Rights This article does not contain any studies with human or animal subjects performed by any of the authors.

Competing Interests The authors declare no competing interests.

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